

Day 3: DNS Server Configuration Theory

Q1. Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in.

Ans:

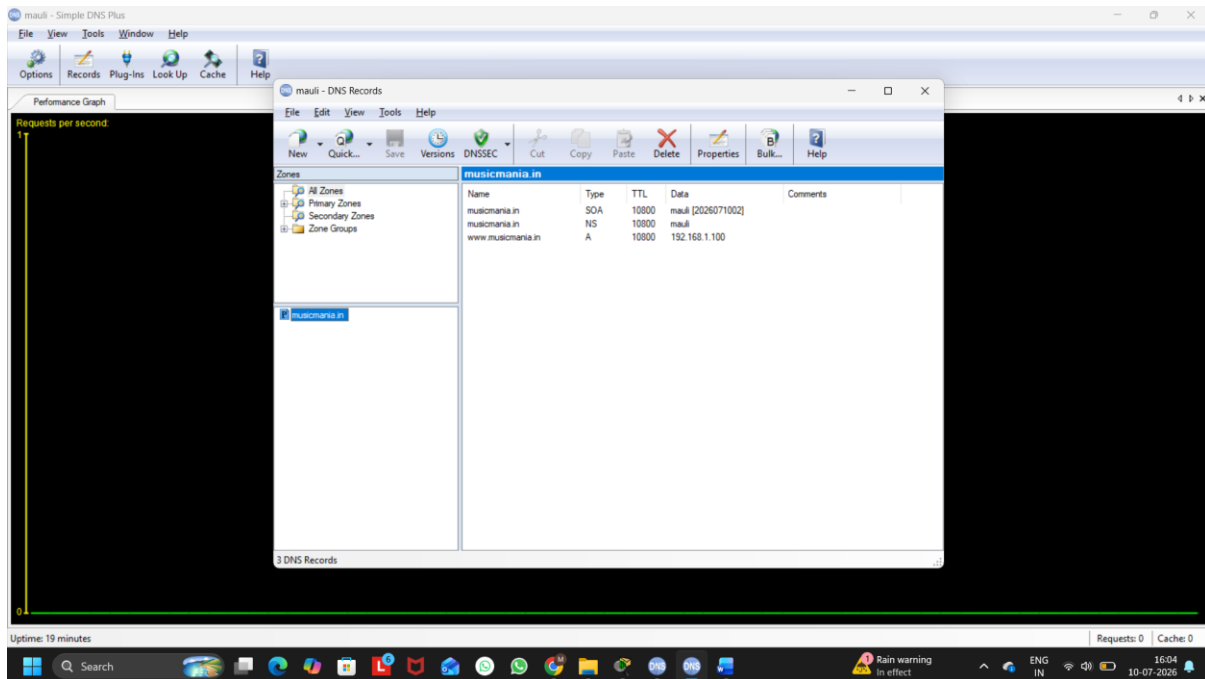


Figure 1: DNS Zone (musicmania.in) with A Record Configuration in Simple DNS Plus

Explanation:

A new DNS Primary Zone named **musicmania.in** was created in Simple DNS Plus. An **A Record** (www.musicmania.in) was added and mapped to the IP address **192.168.1.100**. This allows the DNS server to resolve the domain name to its corresponding IP address.

Q2. Add an A record in your DNS server configuration to map www.musicmania.in to the IP address 192.168.1.50, then use the nslookup or dig command to verify the mapping works.

Ans:

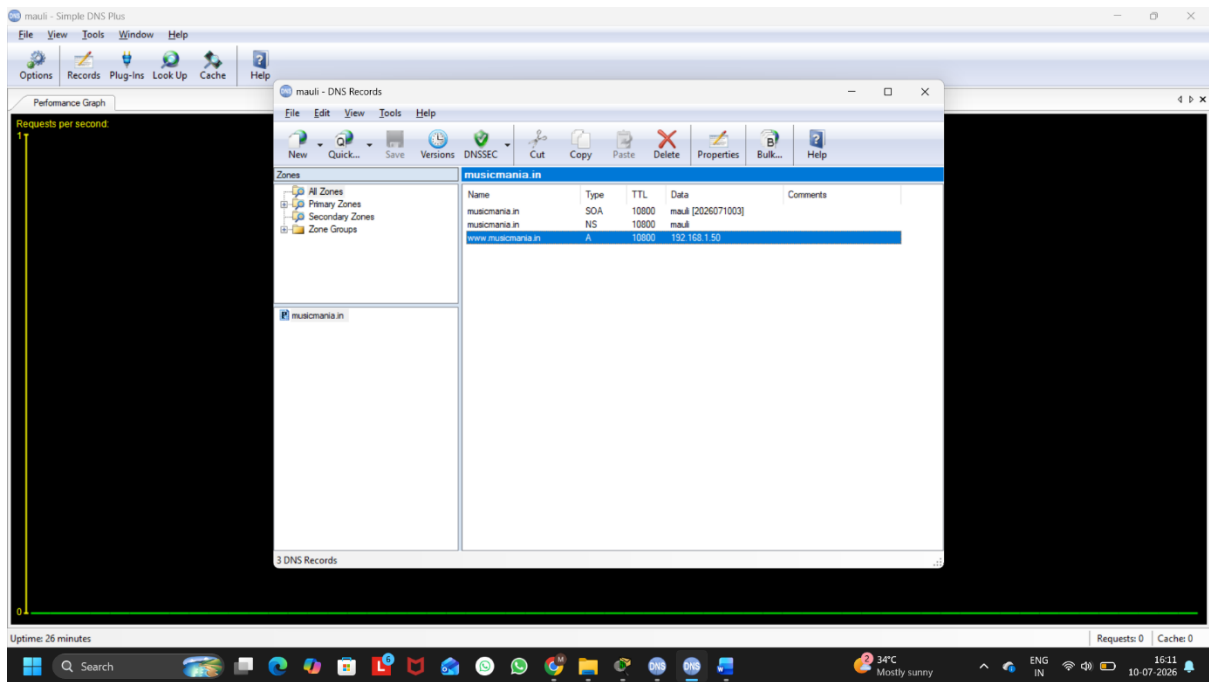


Figure 1: A Record Mapping www.musicmania.in to 192.168.1.50

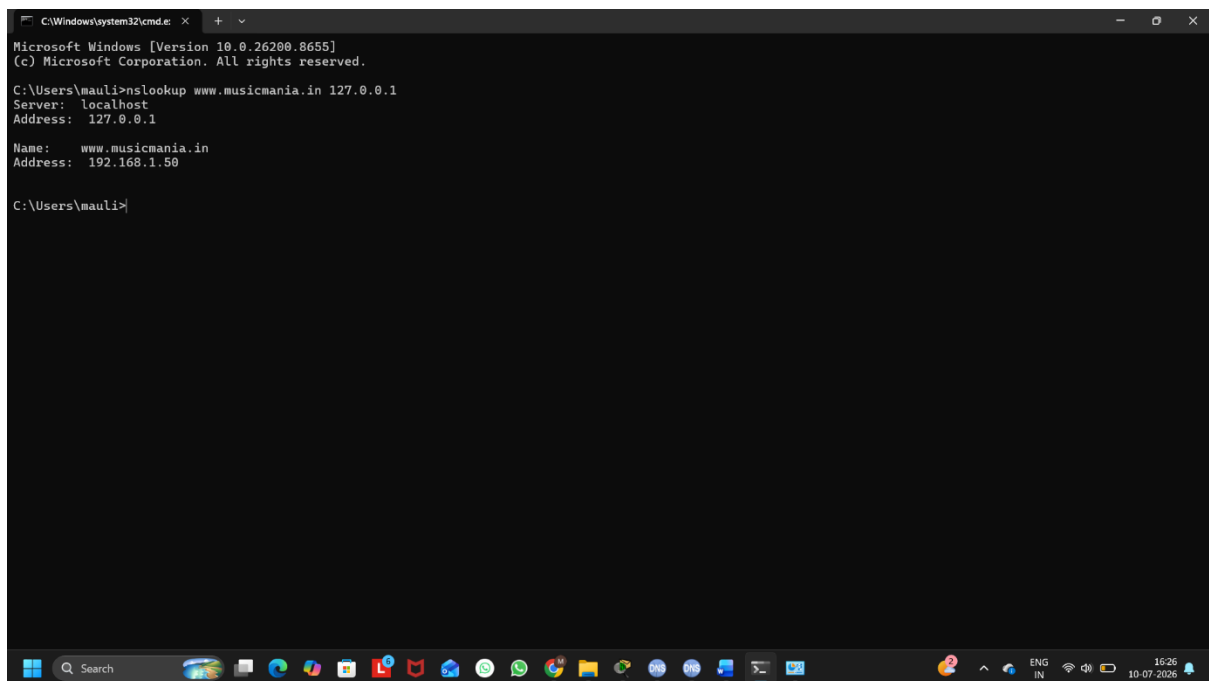


Figure 2: NSLookup Verification of www.musicmania.in

Verification:

The A record was created to map www.musicmania.in to the IP address **192.168.1.50**. The **nslookup** command successfully resolved the domain name to the configured IP address, confirming that the DNS server is working correctly.

Q3. Simulate a typo by adding a CNAME record for play.musicmania.in that points to www.musicmania.in, then test if accessing play.musicmania.in resolves to the correct IP address.

Ans:

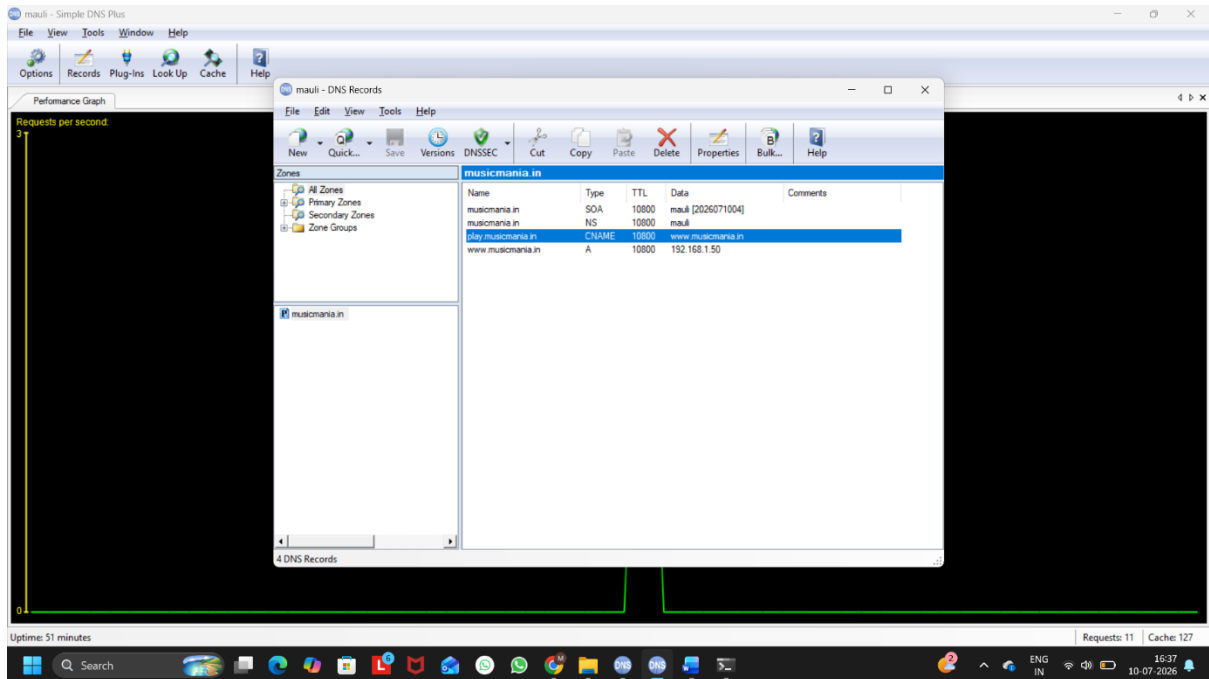


Figure 1: CNAME Record for play.musicmania.in

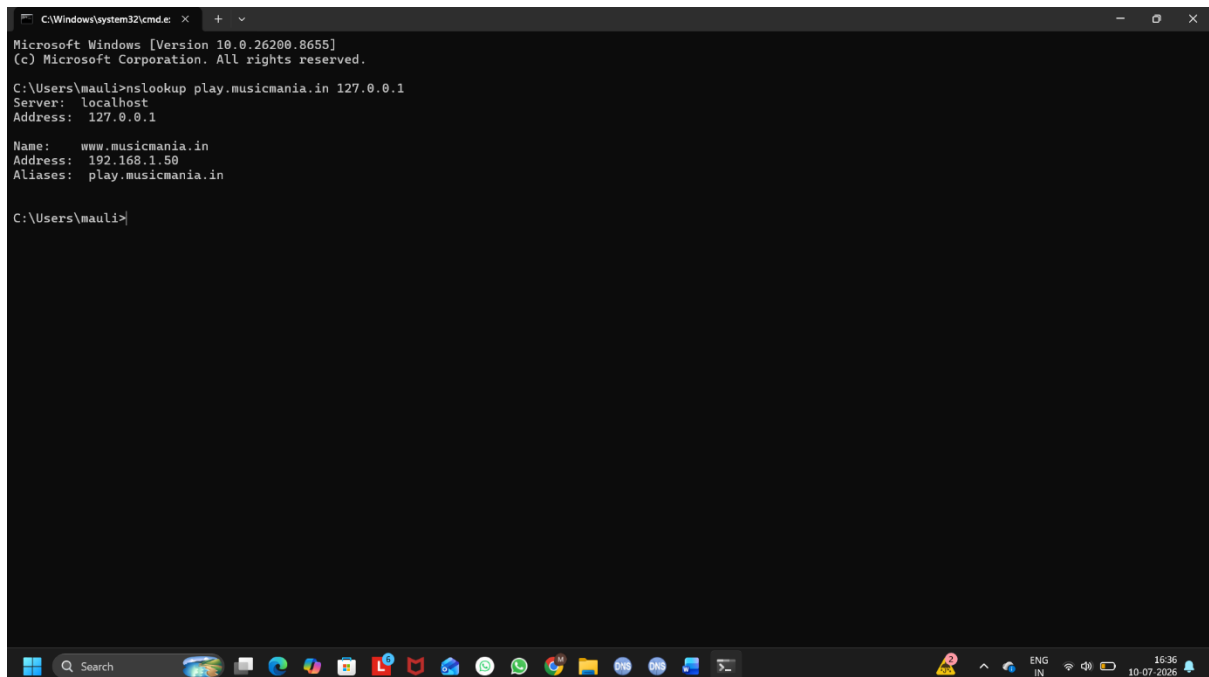


Figure 2: NSLookup Verification of play.musicmania.in

Explanation:

The **CNAME** record maps **play.musicmania.in** to www.musicmania.in instead of directly storing an IP address. When **nslookup** is executed, the DNS server first resolves the alias to www.musicmania.in and then returns the IP address **192.168.1.50**.

**Q4. Your DNS server is not resolving the domain musicmania.in as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with nslookup or dig.

Hint: Include at least three different troubleshooting commands or checks in your documentation.**

Ans:

Step-by-Step Troubleshooting:

Step 1: Check DNS Zone Configuration

- Open Simple DNS Plus.
- Verify that the musicmania.in zone exists.
- Check that the A Record for www.musicmania.in points to the correct IP address (192.168.1.50).
- Ensure there are no spelling mistakes in the domain name.

Step 2: Verify DNS Records

- Confirm that all required DNS records (SOA, NS, A, or CNAME) are present.
- Save any recent changes made to the DNS configuration.

Step 3: Restart the DNS Service

- Open Simple DNS Plus.
- Restart the DNS service to apply configuration changes.
- Wait a few seconds for the service to start again.

Step 4: Test Using nslookup

Open Command Prompt and run:

nslookup www.musicmania.in 127.0.0.1

Expected Result:

Name: www.musicmania.in

Address: 192.168.1.50

Step 5: Test Another Record

- If a CNAME record exists, verify it using:

- nslookup play.musicmania.in 127.0.0.1
- This confirms that the alias correctly resolves to the configured IP address.

Step 6: Check DNS Service Status

Ensure that:

- The DNS service is running.
- The firewall is not blocking DNS requests.
- Port 53 is available.

Commands / Checks Used:

nslookup www.musicmania.in 127.0.0.1

nslookup play.musicmania.in 127.0.0.1

ipconfig /flushdns